

B.Sc (HONS.) IN CSE, FIRST YEAR, FIRST SEMESTER EXAMINATION, 2020
STRUCTURED PROGRAMMING LANGUAGE

[According to the New Syllabus]

Subject Code : CSE-510201

Examination Code : 5611

Time—3 hours

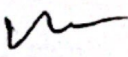
Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | Marks |
|---|-------|
| 1. (a) What do you mean by structured programming language?
Write down some salient features of structured programming language. | 5 |
| (b) Describe the basic structure of a C-program. | 5 |
| (c) Define flowchart. Draw the flowchart to find the largest value among three numbers. | 5 |
| (d) Write down the algorithm to find the area and perimeter of a rectangle. | 5 |
| 2. (a) What is variable? How can you define and initialize a variable in C-program? | 5 |
| (b) Describe with example the four basic data types used in a C-program. | 5 |
| (c) Distinguish between : | 2×2=4 |
| (i) getchar () and scanf () functions | |
| (ii) local variable and global variable | |
| (d) Write a C-program to convert the temperature from Fahrenheit to Celsius and vice versa. Use the following formula : | 6 |

$$\frac{C}{5} = \frac{F - 32}{9}$$

[Please turn over

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- | | Marks |
|---|-------|
| 3. (a) What is looping? Explain entry and exit control loop. | 1+4=5 |
| (b) Write down the general form and flowchart of the following statements : | 5 |
| (i) if else | |
| (ii) for | |
| (c) What is infinite loop? What are the uses of continue and break in C? | 2+3=5 |
| (d) Write a C-program to calculate the sum of following series :
1 + 2+ 3+ 4 + + 50. | 5 |
| 4. (a) What do you mean by linear array? Describe the declaration and initialization of a linear array. | 1+4=5 |
| (b) Define pointer. Write down the advantages and disadvantages of using pointer. | 1+4=5 |
| (c) Write a C-program to read an array and display its elements in reverse order. | 5 |
| (d) Describe the logical and relational operators used in C-language. | 5 |
| 5. (a) What is function? Why do we need to use function in C? | 4 |
| (b) Explain call by value and call by reference with example. | 5 |
| (c) What do you mean by recursion? Write a C-program to find first n Fibonacci numbers using recursion. | 1+5=6 |
| (d) Describe some string handling functions used in C with example. | 5 |
| 6. (a) What is the significant of EOF in file operation? Write down the advantages of file. | 4 |
| (b) What are the purposes of memory management function malloc and calloc? | 6 |
| (c) How can you overcome the limitation of an array using structure? | 4 |
| (d) Write a program to copy the contents of one file to another file. | 6 |